

Group 3 Query Document

Query #1

Count the total number of patients in the database: Calculates the aggregate number of registered patient records inside the clinic's system.

```
2
3  -- Query #1
4  -- Count the total number of patients in the database.
5  -- Written by: Jazmin
6  SELECT COUNT(*) AS total_patients
7  FROM patient;
8
```

Data Output		Messages	Notifications
		SQL	
	total_patients bigint		
1	20		

Query #2

Count how many patients visited the clinic last month: Determines the specific number of patients who successfully attended an appointment during the previous month.

```
8
9  -- Query #2
10 -- Count how many patients visited the clinic last month.
11 -- Written by: Jazmin
12 SELECT COUNT(DISTINCT a.patient_id) AS patients_last_month
13 FROM visit v
14 JOIN appointment a ON v.appointment_id = a.appointment_id
15 WHERE a.date_time >= date_trunc('month', CURRENT_DATE - INTERVAL '1 month')
16       AND a.date_time < date_trunc('month', CURRENT_DATE);
17
18 -- Query #3
19 -- Calculate the total dollar amount billed to all patients.
20 -- Written by: Jazmin
```

Data Output		Messages	Notifications
		SQL	
	patients_last_month bigint		
1	12		

Query #3

Calculate the total dollar amount billed to all patients: Adds the overall financial value of all the invoices created across every visit recorded on file.

```

17
18 -- Query #3
19 -- Calculate the total dollar amount billed to all patients.
20 -- Written by: Jazmin
21 SELECT SUM(total_amount) AS total_billed_amount
22 FROM bill;
23
24
25
26 -- Custom Queries --

```

Data Output

Messages

Notifications

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SQL

	total_billed_amount numeric 🔒
1	7189.00

Query #4

Demonstrate aggregation (SUM, COUNT, AVG): Utilizes aggregate functions to group up and summarize specific operational or financial data of the clinic.

[OBJ:OBJ]

Query #5

Subqueries and JOINS: Employs nested subqueries and relational table joins to extract correlated data across multiple clinic tables.

```
10 -- One query that uses subqueries and JOINS (e.g., INNER, LEFT, RIGHT).
11 -- Dylane
12
13 SELECT CONCAT(pe.first_name, ' ', pe.last_name) AS "Full Name", a.date_time AS "Appointment Date and Time", a.status AS "Status", a.notes AS "Booking Notes", v.diagnosis AS "Diagnosis", t.treatment AS "Treatment", d.notes AS "Doctor's Notes"
14 FROM visit v
15 INNER JOIN appointment a USING (appointment_id)
16 INNER JOIN patient p ON a.patient_id = p.person_id
17 INNER JOIN person pe USING (person_id)
18 WHERE EXTRACT(MONTH FROM a.date_time) = EXTRACT(MONTH FROM NOW())
19 AND a.patient_id IN (
20     SELECT patient_id
21     FROM parent_guardian
22 )
23
24
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	Full Name text	Appointment Date and Time timestamp with time zone	Status character varying (20)	Booking Notes character varying (500)	Diagnosis character varying (255)	Treatment character varying (255)	Doctor's Notes character varying (500)
1	Maya Martin	2026-09-01 09:15:00-06	ATTENDED	Fever and sore throat	Strep throat	Rapid strep test positive; antibiotics prescri...	Stay home until 24h after first d...
2	Jack Benn...	2026-09-02 15:30:00-06	ATTENDED	Ear infection follow-up	Otitis media, resolved	Ear exam; infection cleared	[null]
3	Leo Singh	2026-09-24 15:45:00-06	ATTENDED	Asthma follow-up	Mild persistent asthma, sta...	Dose reviewed after August adjustment	[null]

Query #6

GROUP BY and HAVING: Identifies patient–doctor pairs where the patient has had at least five visits with that doctor.

Query

Query History

1

-- One query that uses GROUP BY with HAVING.

2

-- Select all patients, group by doctor having visits >= 5

3

SELECT

4

p.first_name || ' ' || p.last_name AS patient_name,

5

d.first_name || ' ' || d.last_name AS doctor_name,

6

count(*) AS total_visits

7

FROM visit v

8

JOIN appointment a ON a.appointment_id = v.appointment_id

9

JOIN person p ON p.person_id = a.patient_id

10

JOIN person d ON d.person_id = a.doctor_id

11

GROUP BY

12

patient_name,

13

doctor_name

14

HAVING count(*) >= 5

15

ORDER BY total_visits DESC;

16

Data Output

Messages

Notifications

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SQL

	patient_name text	doctor_name text	total_visits bigint
1	Lucas Fraser	Sofia Rossi	8
2	Leo Singh	Emily Nguyen	7
3	Jack Bennett	Emily Nguyen	6
4	Ethan Roy	Amara Okafor	6
5	Evelyn Clarke	Sofia Rossi	6
6	Noah Campbell	Liam Chen	5
7	Olivia Bennett	Amara Okafor	5

Query #7

Stored view: Retrieves the date, patient, doctor, and notes for appointments marked as follow-up appointments.

```

16
17 -- One query that uses a stored procedure or view.
18 -- Get appointment date, patient name, doctor name, notes
19 CREATE VIEW follow_up_appointment AS
20 SELECT
21     p.first_name || ' ' || p.last_name AS patient_name,
22     d.first_name || ' ' || d.last_name AS doctor_name,
23     a.date_time,
24     a.notes
25 FROM appointment a
26 JOIN person p ON p.person_id = a.patient_id
27 JOIN person d ON d.person_id = a.doctor_id
28 WHERE a.notes ILIKE '%follow-up%';
29 s|
30 SELECT * FROM follow_up_appointment;
31

```

Data Output Messages Notifications

SQL

	patient_name text	doctor_name text	date_time timestamp with time zone	notes character varying (500)
1	Ava Singh	Priya Sharma	2026-08-05 08:45:00-06	Follow-up on blood work
2	Olivia Bennett	Amara Okafor	2026-09-08 10:00:00-06	Follow-up on lab results
3	Ethan Roy	Amara Okafor	2026-09-15 11:00:00-06	Back pain follow-up
4	Ava Singh	Priya Sharma	2026-10-06 09:00:00-06	Follow-up appointment
5	Leo Singh	Emily Nguyen	2026-02-26 08:30:00-07	Asthma follow-up
6	Evelyn Clarke	Sofia Rossi	2026-04-02 13:30:00-06	Cholesterol follow-up
7	Lucas Fraser	Sofia Rossi	2026-04-07 09:30:00-06	Blood pressure follow-up
8	Noah Campbell	Liam Chen	2026-05-05 10:00:00-06	Heartburn follow-up
9	Leo Singh	Emily Nguyen	2026-05-07 08:30:00-06	Asthma follow-up
10	Amelia Patel	Priya Sharma	2026-05-12 10:30:00-06	Asthma follow-up
11	Ethan Roy	Marcus Johnson	2026-05-26 13:00:00-06	Back pain follow-up
12	Jack Bennett	Emily Nguyen	2026-09-02 15:30:00-06	Ear infection follow-up
13	Evelyn Clarke	Sofia Rossi	2026-09-22 13:30:00-06	Cholesterol follow-up
14	Isabella Martin	Daniel Tremblay	2026-09-23 13:30:00-06	Rash follow-up
15	Charlotte Dubois	Hannah Kowal...	2026-09-23 15:45:00-06	Migraine follow-up
16	Lucas Fraser	Sofia Rossi	2026-09-24 09:30:00-06	Blood pressure follow-up
17	Leo Singh	Emily Nguyen	2026-09-24 15:45:00-06	Asthma follow-up
18	James Sutherla...	Marcus Johnson	2026-09-25 14:00:00-06	Ankle follow-up

Query #8

CTEs and window functions (ranking): For each doctor, identifies the top two patients with the highest total billed amount.

```

82
83 -- Query #8
84 -- For each doctor, who are their two highest-billed patients?
85 -- Written by: Umayya
86 WITH patient_bills AS (
87     SELECT a.doctor_id,
88            a.patient_id,
89            p.first_name || ' ' || p.last_name AS patient_name,
90            SUM(b.total_amount) AS total_billed
91     FROM visit v
92     JOIN appointment a USING (appointment_id)
93     JOIN bill b USING (visit_id)
94     JOIN patient pa ON pa.person_id = a.patient_id
95     JOIN person p ON p.person_id = pa.person_id
96     GROUP BY a.doctor_id,
97            a.patient_id,
98            patient_name
99 ),
100 ranked AS (
101     SELECT doctor_id,
102            p.first_name || ' ' || p.last_name AS doctor_name,
103            patient_name,
104            total_billed,
105            row_number() OVER (PARTITION BY doctor_id ORDER BY total_billed DESC) AS billed_rank
106     FROM patient_bills
107     JOIN doctor d ON d.person_id = doctor_id
108     JOIN person p ON p.person_id = d.person_id
109 )
110 SELECT doctor_name, billed_rank, patient_name, total_billed
111 FROM ranked
112 WHERE billed_rank <= 2
113 ORDER BY doctor_name, billed_rank;
114

```

Data Output Messages Notifications

	doctor_name text	billed_rank bigint	patient_name text	total_billed numeric
1	Amara Okafor	1	Ethan Roy	420.00
2	Amara Okafor	2	Olivia Bennett	405.00
3	Daniel Tremblay	1	Henry Kim	235.00
4	Daniel Tremblay	2	Isabella Martin	185.00
5	Emily Nguyen	1	Leo Singh	589.00
6	Emily Nguyen	2	Jack Bennett	494.00
7	Hannah Kowalski	1	Charlotte Dubois	280.00
8	Hannah Kowalski	2	Olivia Bennett	150.00
9	Liam Chen	1	Noah Campbell	450.00
10	Liam Chen	2	Benjamin Wong	150.00
11	Marcus Johnson	1	Ethan Roy	175.00
12	Marcus Johnson	2	James Sutherla...	172.50
13	Omar Haddad	1	William Murray	185.00
14	Priya Sharma	1	Amelia Patel	380.00
15	Priya Sharma	2	Ava Singh	238.50
16	Robert MacDon...	1	Mia Gagnon	195.00
17	Robert MacDon...	2	Olivia Bennett	160.00
18	Sofia Rossi	1	Lucas Fraser	575.00
19	Sofia Rossi	2	Evelyn Clarke	515.00

Query #9

CTEs and window functions (month-to-month comparison): Displays the monthly change and percentage difference in the number of visits and total billing.

```

115 -- Query #9
116 -- How did visits and billing change from month to month?
117 -- Written by: Umayya
118 WITH monthly AS (
119     SELECT date_trunc('month', a.date_time)::date AS month,
120            COUNT(v.visit_id) AS total_visits,
121            SUM(b.total_amount) AS total_billed
122     FROM visit v
123     JOIN appointment a USING (appointment_id)
124     JOIN bill b USING (visit_id)
125     GROUP BY 1
126 ),
127 previous_month AS (
128     SELECT month,
129            total_visits,
130            total_billed,
131            lag(total_visits) OVER (ORDER BY month) AS previous_visits,
132            lag(total_billed) OVER (ORDER BY month) AS previous_billed
133     FROM monthly
134 ),
135 monthly_changes AS (
136     SELECT month,
137            total_visits,
138            total_billed,
139            previous_visits,
140            previous_billed,
141            total_visits - previous_visits AS visits_delta,
142            total_billed - previous_billed AS billed_delta
143     FROM previous_month
144 )
145 SELECT month,
146        total_visits,
147        round(total_billed, 2) AS total_billed,
148        visits_delta,
149        round(100.0 * (visits_delta / NULLIF(previous_visits, 0)), 1) AS mom_visits_pct,
150        round(billed_delta, 2) AS billed_delta,
151        round(100.0 * (billed_delta / NULLIF(previous_billed, 0)), 1) AS mom_billed_pct
152 FROM monthly_changes
153 ORDER BY month;
154

```

Data Output Messages Notifications

	month date	total_visits bigint	total_billed numeric	visits_delta bigint	mom_visits_pct numeric	billed_delta numeric	mom_billed_pct numeric
1	2026-01-...	6	597.00	[null]	[null]	[null]	[null]
2	2026-02-...	7	550.00	1	0.0	-47.00	-7.9
3	2026-03-...	7	577.00	0	0.0	27.00	4.9
4	2026-04-...	7	655.00	0	0.0	78.00	13.5
5	2026-05-...	8	717.00	1	0.0	62.00	9.5
6	2026-06-...	7	730.00	-1	0.0	13.00	1.8
7	2026-07-...	1	85.00	-6	0.0	-645.00	-88.4
8	2026-08-...	22	2238.00	21	2100.0	2153.00	2532.9
9	2026-09-...	15	1040.00	-7	0.0	-1198.00	-53.5

Query #10

CTEs and window functions (average and percentage comparison): Identifies doctors who bill more than the average doctor, and shows the percentage of total clinic billing contributed by each doctor.

```

155 -- Query #10
156 -- Which doctors bill more than the average doctor, and what share of all clinic billing is each one's?
157 -- Written by: Umayya
158 WITH doctor_billings AS (
159     SELECT a.doctor_id,
160            p.first_name || ' ' || p.last_name AS doctor_name,
161            SUM(b.total_amount) AS total_billed
162     FROM visit v
163     JOIN appointment a USING (appointment_id)
164     JOIN bill b USING (visit_id)
165     JOIN doctor d ON d.person_id = a.doctor_id
166     JOIN person p ON p.person_id = d.person_id
167     GROUP BY a.doctor_id,
168              doctor_name
169 ),
170 billing_stats AS (
171     SELECT doctor_id,
172            doctor_name,
173            total_billed,
174            round(AVG(total_billed) OVER (), 2) AS avg_doctor_billing,
175            SUM(total_billed) OVER () AS total_clinic_billing
176     FROM doctor_billings
177 )
178 SELECT doctor_name,
179        total_billed,
180        avg_doctor_billing,
181        total_clinic_billing,
182        round(100.0 * (total_billed / NULLIF(total_clinic_billing, 0)), 1) AS pct_of_clinic_billing
183 FROM billing_stats
184 WHERE total_billed > avg_doctor_billing
185 ORDER BY pct_of_clinic_billing DESC;

```

Data Output Messages Notifications

	doctor_name text	total_billed numeric	avg_doctor_billing numeric	total_clinic_billing numeric	pct_of_clinic_billing numeric
1	Emily Nguyen	1723.00	718.90	7189.00	24.0
2	Sofia Rossi	1090.00	718.90	7189.00	15.2
3	Priya Sharma	953.50	718.90	7189.00	13.3
4	Amara Okafor	825.00	718.90	7189.00	11.5

Contribution Table

Team Member	Roles / Tasks
Justin Shaw	Created the schema and custom queries
Dylane Cunningham	Created the EERD and custom queries
Jazmin Horton	Did all required queries and documentation
Umayya Hewaarachchi	Custom queries and documentation

